

Discrete Random Variables & the Binomial Distribution



Probability distributions and expected value

- 1 A game costs 2 dollars to play. You win 10 dollars with probability 0.1, and win nothing otherwise. Let X be your net profit. Find the probability distribution of X and its expected value.
 - 2 Find the standard deviation of the profit X from the previous problem, to two decimal places.
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Binomial setting and conditions

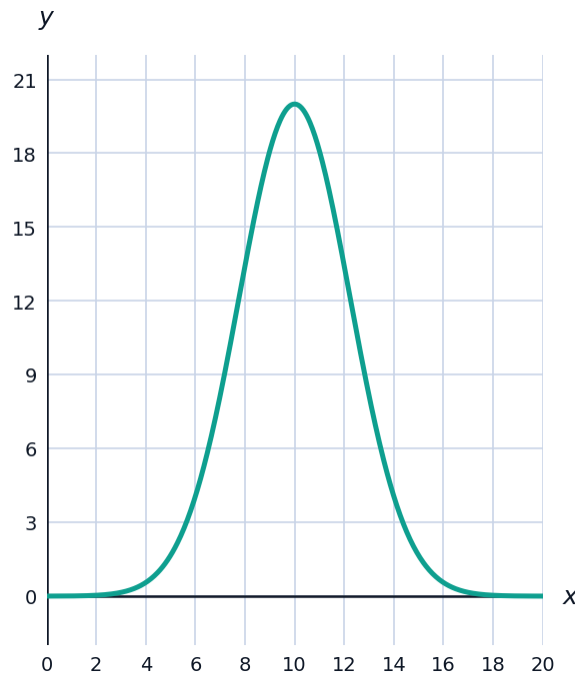
- 3 State the four conditions required for a random variable to follow a binomial distribution (sometimes remembered as BINS: Binary, Independent, Number fixed, Same probability).
 - 4 A survey randomly selects 20 people (without replacement) from a small town of 50 people and records how many support a proposal. Explain why this technically violates the independence condition, and why it is often still treated as approximately binomial.
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Binomial probability formula

- 5 A basketball player makes 70% of free throws. Find the probability she makes exactly 4 out of 6 attempts, using $P(X = k) = \binom{n}{k} p^k (1 - p)^{n - k}$.
 - 6 Using the same free-throw shooter, find the mean and standard deviation of the number of makes out of 6 attempts, using $\mu = np$ and $\sigma = \sqrt{np(1 - p)}$.
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Visualizing a binomial distribution

- 7 The graph shows a smoothed approximation of the binomial distribution for $n = 20$ trials with $p = 0.5$, plotted as a continuous curve for visual comparison to a normal shape.



- a Using $\mu = np$ and $\sigma = \sqrt{np(1-p)}$, verify that this binomial distribution has mean 10 and standard deviation approximately 2.24, matching the curve shown.
- b Explain why a binomial distribution with $n = 20$ and $p = 0.5$ looks approximately bell-shaped, referencing the conditions for a normal approximation.

Geometric distribution (brief introduction)

- 8 A basketball player makes free throws with probability 0.7 each attempt, independently. Find the probability that her first miss occurs on the 3rd attempt (i.e. she makes the first 2, then misses the 3rd).

Applying the binomial distribution in context

- 9 A multiple-choice quiz has 10 questions, each with 4 choices, and a student guesses randomly on every question. Find the probability the student gets at least 8 questions correct, given that $P(X = 8) \approx 0.0004$, $P(X = 9) \approx 0.00003$, and $P(X = 10) \approx 0.000001$.

Binomial probability of at least one success

- 10 A manufacturing process produces defective items independently with probability 0.05. In a batch of 20 items, find the probability that at least one item is defective, using the complement of "zero defective."

- 11 Using the mean and standard deviation formulas $\mu = np$ and $\sigma = \sqrt{np(1 - p)}$, find the mean and standard deviation of the number of defective items in a batch of 20, with $p = 0.05$.
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Comparing binomial scenarios

- 12 Explain why a binomial model would NOT be appropriate for counting the number of aces dealt when drawing 5 cards without replacement from a standard 52-card deck, and name the distribution that would be more appropriate instead.